Wireshark Lab 1: Introduction to Wireshark

Objective

To familiarize students with the Wireshark interface, understand basic packet capturing, and learn how to apply filters for protocol analysis.

Requirements

- Wireshark installed on your system.
- A working internet connection.
- Administrative/root privileges for packet capturing.

Step-by-Step Instructions

Step 1: Launch Wireshark

1. Open Wireshark from the Start Menu (Windows) or Applications (Linux/Mac).
2. You will see a list of network interfaces (e.g., Wi-Fi, Ethernet, Loopback).

Step 2: Start a Packet Capture

1. Select your active network interface (e.g., Wi-Fi if you are connected wirelessly).
2. Click the blue shark fin icon at the top.
3. Wireshark will now display live network traffic in the Packet List Panel.

Step 3: Explore the Wireshark Interface

Wireshark's window has three important panels:
- Packet List Panel – shows summary info for each packet (time, source, destination, protocol, length).
- Packet Details Panel – shows detailed breakdown of the selected packet (Ethernet, IP, TCP/UDP, etc.).
- Packet Bytes Panel – displays the raw packet data in hexadecimal and ASCII.

Step 4: Stop and Save the Capture

1. Click the red square icon to stop capturing.
2. Go to File > Save As.
3. Save your capture as Lab1-Intro.pcapng.

Step 5: Apply Display Filters

In the display filter bar, type:
- ip – shows only IP packets.
- tcp – shows only TCP packets.
- udp – shows only UDP packets.
- http – shows only HTTP packets (if browsing was done).
Press Enter to apply each filter.

Step 6: Inspect a Packet

1. Select any packet from the Packet List.
2. Expand protocol layers (Ethernet -> IP -> TCP/UDP -> Application).
3. Note down:
   - Source IP address
   - Destination IP address
   - Protocol used

Wireshark Task 2: Capturing Live Network Traffic

Objective

Learn how to start, stop, and save live network traffic captures using Wireshark.

Step-by-Step Instructions

Step 1: Select Network Interface

1. Open Wireshark.
2. Choose the appropriate interface from the list (e.g., Wi-Fi or Ethernet).
3. Look for the interface with active packet counts.

Step 2: Start Capturing Packets

1. Click the **blue shark fin** icon to start the capture.
2. Open a browser and visit any website (e.g., *www.example.com*).
3. Watch the packets being captured in real time.

Step 3: Observe Live Traffic

1. Note the rapidly updating packet list.
2. Identify different protocols such as **TCP, UDP, DNS, HTTP, etc.**

3. Use the display filter (top bar) to narrow the visible traffic, for example:
    - http
    - dns
    - tcp.port == 80

Step 4: Stop the Capture

1. Click the **red square** icon to stop the capture.
2. Make sure the captured traffic appears in the list and can be scrolled.

Step 5: Save the Capture

1. Go to **File > Save As**.
2. Save the file as **Lab2-Capture.pcapng**.
3. Choose a known location (like Desktop or Documents).

Step 6: Reopen Saved Capture

1. Close and reopen Wireshark.
2. Go to **File > Open** and load **Lab2-Capture.pcapng**.
3. Verify that all packets were saved.

Wireshark Lab 3: Analyzing Ethernet Frames

Objective:

Understand the structure and content of Ethernet frames using Wireshark. Learn to identify MAC addresses and Ethernet frame types.

Requirements:

- Wireshark installed and functional.
- Two devices on the same LAN (optional but useful for ARP broadcasts).
- Ping command access or any web browser.

Step-by-Step Instructions:

Step 1: Start Wireshark and Begin Capture

1. Launch Wireshark.
2. Select the active network interface (e.g., Wi-Fi or Ethernet).
3. Click the **blue shark fin** to start capturing.

Step 2: Generate Traffic

1. Open a terminal or command prompt.
2. Type ping www.example.com or visit any website in your browser.
3. Let it run for 4–5 pings, then stop the command or close the browser tab.

Step 3: Stop the Capture

1. Return to Wireshark.
2. Click the **red square icon** to stop capturing.
3. Use the filter bar and enter:

   ```
   eth
   ```

   to view only Ethernet frames.

Step 4: Analyze Ethernet Frame Structure

1. Click on a packet from the list.
2. In the **Packet Details Panel**, expand the **Ethernet II** section.
3. Observe the following:
   - **Destination MAC Address**: e.g., b8:27:eb:12:34:56
   - **Source MAC Address**: e.g., d0:50:99:89:ab:cd
   - **Type**: Indicates the protocol encapsulated (e.g., IPv4, ARP).

Step 5: Filter ARP Packets (Optional)

1. In the filter bar, type:

   ```
   arp
   ```

2. Find packets with Who has and is at messages.
3. These are Ethernet broadcast frames used for address resolution.

Step 6: Save Your Capture

1. Go to **File > Save As**.
2. Save the file as Lab3-Ethernet.pcapng.